Deep Learning Model Based CO2 Emissions Prediction Using Vehicle Telematics Sensors Data

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***Abstract*—Climate change is one of the greatest environmental hazards to mankind. The emission of greenhouse gases has resulted in a continuous increase in the temperature of the atmosphere leading to Global warming. CO2 continues to be the leading contributor to the greenhouse effect, with transport being a major CO2 emission source. The majority of transport emissions are from road transport i.e. vehicular emissions. To control vehicular emission, first, an efficient emission monitoring system is required. Direct sensor installation in individual vehicles is neither cost-effective nor the data is easy to collect. In this paper, a scalable vehicle CO2 emission prediction model is proposed which uses vehicle On-Board Diagnostics (OBD-II) port data. The proposed system uses real-time in-vehicle sensor data to estimate CO2 emission of the vehicle using a Recurrent neural network (RNN) based Long short-term memory(LSTM) model. OBD-II dongles can be used to easily transmit the vehicle's sensor data to the cloud, where the LSTM model uses this data to estimate the real-time CO2 emission of the vehicle. The proposed model provides a scalable and efficient system to monitor emissions at a vehicular level and, has been evaluated using public OBD-II dataset. The details of the data collection, sensors, and adapters used along with vehicle information are outlined in Section V-A.**

***Index Terms*—Vehicle telematics, LSTM, RNN, deep learning, OBD-II, CO2 estimation.**

I. Introduction

The climate change continues to haunt mankind as one of the biggest environmental issues of today [1]. Emission of gases like Carbon Dioxide, Nitrous Oxide, and Methane has caused global temperature to increase by 1 °C since before the industrial revolution [2]. CO2 is the leading contributor to the greenhouse effect, with it being the key emission of power, transport, and manufacturing sector. Transport has been a key contributor to CO2 emissions for a long [3]. In India, transport makes up 23.6% of total CO2 emissions, and 65.4% of this is from road transport [4], [5]. Fig. 1 illustrates the key CO2 emission contributors in India as of August 2018.

To control or limit emissions, an efficient and large scale emission monitoring system is required. Vehicular emissions are not easy to monitor because of the sheer number and types of vehicles. Further individual testing of such a large number of vehicles is not feasible. A data driven model has been shown to perform well for CO2 estimation as shown by authors in [6]–[9]. For efficient monitoring of vehicular emissions, a vehicle on-board Diagnostic (OBD) based LSTM model is proposed in this paper. The model uses IoT technology-based dongles to transmit real-time vehicle diagnostic and sensor data to the cloud [10]. An LSTM based model then predicts the CO2 emissions from this data in real-time. Thus, a real-time Vehicular CO2 emission prediction system has been proposed in this paper.

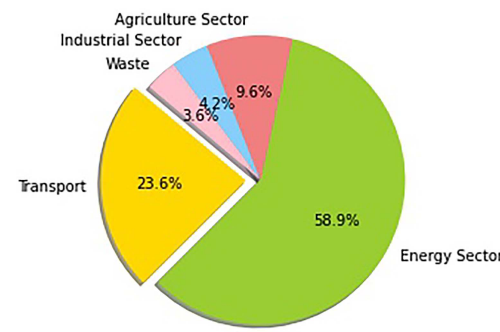

Fig. 1. Share of transport in India's CO2 emission.

The proposed model has been evaluated and compared with other techniques in the literature using an Open Source OBD-II dataset [11]. The data used has been collected and compiled by P. Rettore *et al.* [12]. The data composition and collection have been briefly explained in Section V-A. OBD-II provides in-vehicle sensor readings such as Engine Load, Manifold Pressure, Fuel Trim, and Engine RPM along with vehicle fault codes. IoT based adapters can be used to analyze this data in real-time by sending it to the cloud platform [10], where the predictive model can be run on the data. The emission of any Internal Combustion Engine is directly related to its sensor readings like RPM, load, throttle, etc. Thus, the OBD-II data directly or indirectly reflects knowledge on a vehicle's emission characteristics [13].

Vehicle Telematics data when combined with appropriate machine learning techniques can be designed to predict non-trivial quantities like CO2 emissions, State of Health of Transmission and Drive Train [14], Driving Pattern of Driver [15], Dangerous Driving [16], Vehicle Reaction Time, etc. Such insights into the vehicle allow an efficient real-time health monitoring of the vehicle [14]. Since the telematics data is in the form of a time series, traditional machine learning techniques cannot be used

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directly. Traditional machine learning algorithms, which train on only the current example without a memory of past results are not able to analyze sequential relationships in the data, hence, are not ideal for time series prediction.

A Long Short-Term Memory (LSTM) [17] based deep learning architecture has been proposed in this study. LSTM is a Recurrent Neural Network (RNN) [18] variant which remembers previous inputs along with the current example, hence, is able to exploit the sequential relationships in data effectively. Since, deep learning models generate features on their own, rigorous feature engineering and extraction is not required by the model. The model has been evaluated against noisy data to simulate real-life sensor perturbations.

Section II presents a literature review of existing methods applied to the estimation of CO2 from mobile platforms. Section III gives a theoretical background of the techniques used. Section IV outlines the implementation details. Sections V and Section VI summarize the experimentation and results. Finally, the contributions of the study are highlighted in Section VII along with limitations and future works.

II. Related Work

The generalized deployment of cellular communications technologies implies that telematic applications can be highly accessible [10]. This is the primary reason for the popularity of telematics-based prediction systems. Further, with advancements in machine learning and big data, telematics can provide very valuable insights into the systems. A brief review of some of the work done follows:

The authors in [6] have used Artificial Neural Network (ANN), Support Vector Machine (SVM), and VT-Micro techniques for CO2 estimation. Further, they have used speed, acceleration and throttle data collected via the On-Board Diagnostics (OBD) port at intervals of 30 seconds to find the environmentally optimal route for vehicles. The issues outlined by the authors was the poor estimation of CO2 emission with only 2 features. The future work of the study was stated to be a large scale emission estimation system. In this work, we have attempted to propose a large scale vehicular emission estimation technique.

In [7], authors have used data generated by Inductive Loop Detectors (ILD) provided by Urban Transport Corporation (UTC). Emission models were designed based on this data. Further, they have also integrated vehicle sensor data namely speed, acceleration, fuel flow and mileage with their model. These readings are collected via the OBD port at intervals of 30 seconds. The estimation is based on a hybrid model combining the OBD based model and the ILD model. The OBD model is used to set the parameters for the ILD model. The results obtained, show that ILD data along with categorization of vehicle types through OBD model can form the basis for a practical road traffic CO2 modeling.

In [8], authors have done a regressive analysis of CO2 prediction using speed and acceleration of the vehicle. The results obtained by them show a linear relationship between CO2, speed, and acceleration/deceleration of the vehicle. Speed as compared to acceleration is found to have a stronger correlation with respect to the CO2 emission factor. One of the main ideas in the paper is to use a wider set of OBD features. This is exactly what this study does.

In [9], authors studied the effects of the distance of trip on the fuel consumption of the vehicle. They employed deep learning techniques to predict fuel consumption for some fixed routes. The authors only used speed and distance values collected at intervals of 60 seconds. The non-linear relationship found by the study shows that the fuel consumption depends on many factors other than speed and distance of the vehicle. The authors show a correlation between speed and fuel flow. The future work for this study was to use a wide set of OBD features to accurately model fuel flow which is exactly what our study proposes. The correlation found can be extended to CO2 estimation too.

In their journal paper titled, “The value of vehicle telematics data in insurance risk selection processes” [19], authors have used vehicle telematics data to calculate a risk metric of an individual, to be weighed in while calculating insurance risk. Three machine-learning based regression techniques have been used namely, Deep Neural Network, Logistic Regression, and Random Forest Estimator. The study shows the importance of telematics data in the insurance industry and the knowledge that the data contains.

The authors in [15] and [20] have proposed a sequence classification model for driver identification from OBD-II vehicle telematics data. An LSTM model has been proposed in the study which obtains an accuracy of 97% showing that the vehicle telematics data reflects driving style. The study also shows that telematics data can be used to identify risky and aggressive driving patterns. The robustness of the model has also been tested using noisy and anomalous data.

The literature review shows that the current CO2 estimation and modeling methods rely mainly on speed, congestion data along with specialized sensors. Speed and acceleration, though highly co-related with CO2 emission, can only provide limited information into a vehicle’s emission characteristics. Special sensors for CO2 modeling also may be accurate but not very scalable. Thus, these are not suitable for large scale applications. Further, the methods of CO2 estimation in the literature are not able to monitor individual vehicle emissions effectively. Finally, telematics-based prediction systems were considered in the literature like insurance risk prediction, driver identification, etc. The promising results of these studies show that OBD-II can effective in vehicular CO2 estimation. The survey also shows that traditional techniques like SVM, ANN, etc. generally do not perform as well as RNN based approaches for time-based predictions.

The proposed model uses Open Source OBD-II data to predict instantaneous emission values for vehicles. The system takes into account diverse parameters like RPM, Intake Pressure, Load, etc. for this prediction. Further, a detailed feature analysis has been performed to find the optimal subset, thus, removing and irrelevant or redundant data. Finally, RNN based LSTM is used which performs better than ANN and SVM for time-based prediction.

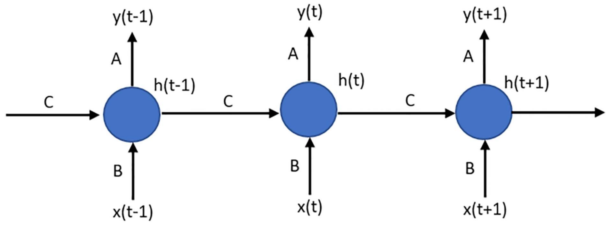

Fig. 2. Structure of an unrolled RNN.

III. Theoretical Background

This section outlines the theory behind the various techniques discussed in this paper. The section covers a brief description of the RNN, LSTM, Vehicle Telematics and, OBD data.

A. Recurrent Neural Network (RNN)

DNN learn only from one training example at a time. Thus, they are not able to take into account the sequential relationship among data [21]. Recurrent Neural Networks were designed to overcome this flaw in DNN. Instead of the gradient depending only on one training example, an RNN takes into account all previous values of the sequence.

For data at instant t, x_t, the value of internal state parameter h_t depends on both the current value x_t and the previous state h_{t-1}. Thus for a general system, the formula for the current state can be written as

$$h_t = f(h_{t-1}, x_t) \tag{1}$$

Here, h_t is the new state, h_{t-1} is the previous state while x_t is the current input. The same weights are applied for all inputs in the sequence. Fig. 2 shows the RNN network in the unrolled form. It can be seen how previous values are transmitted among nodes. RNN suffers from a problem of vanishing gradients, where the weights start to diminish as the training continues. To overcome this, LSTM was introduced.

B. Long Short-Term Memory (LSTM)

RNN passes the information of all nodes in a sequence. This causes RNN gradients to vanish due to repeated sigmoid activations. LSTM also forgets information of previsus nodes making it more robust. LSTM, like RNN, also keeps a memory of the current system state which depends on the previous inputs up till that point, but, LSTM has a unique gate structure controlling the information that is conserved as the network is trained [21], [22]. Fig. 3 shows the architecture of an LSTM cell. c_t represents the internal state of the cell with time t input. x_t represents the input at time x and h_t represents the output of the previous cell.

- *Internal State* - This is the information channel of the LSTM. It carries information from cell to cell. Information is constantly added and removed from the internal state via gates. This ensures that only the most relevant information is carried. It is denoted by c^t.
- *Forget gate* - First the information to be discarded is decided. This is done by a sigmoid function which takes input x_t and h_{t-1} and for each state value in internal state outputs a value between 0 and 1. The value is proportional to the information to be retained. This is called the forget gate, f^t

$$f^{(t)} = \sigma(W^{fX}x^{(t)} + W^{fh}h^{(t-1)} + b_f) \tag{2}$$

- *Input Gate and Node* - New information must also be added to the internal state. This is done by an input gate i^t which uses a sigmoid layer to decide the amount of current data to be added and an input node g^t which creates a set of new internal state values.

$$i^{(t)} = \sigma(W^{iX}x^{(t)} + W^{ih}h^{(t-1)} + b_i) \tag{3}$$

$$g^{(t)} = \tanh(W^{gX}x^{(t)} + W^{gh}h^{(t-1)} + b_g) \tag{4}$$

- *Output Gate* - Finally, for the output ^{t}, a sigmoid layer decides the information to be output. The current state is activated and multiplied to the output gate value, to give the cell output.

$$o^{(t)} = \sigma\left(W^{oX}x^{(t)} + W^{oh}h^{(t-1)} + b_o\right)$$

$$h^{(t)} = \tanh\left(s^{(t)}\right)^{*} o^{(t)} \tag{5}$$

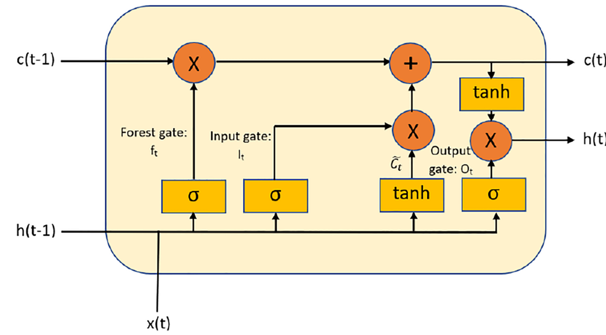

Fig. 3. Structure of a single LSTM cell with input x_t, h_{t-1} and output h_t.

In equations 2–5, W represents the model weights and b represents the model biases. C denotes the cell, h denotes the output and t denotes the time stamps. Thus, LSTM carries useful information for long and discards useless information quickly.

C. Obd-Ii

OBD-II specifications have been recognized by various national and international organizations [23]. On board diagnostics, OBD-II, is a "*higher-layer-protocol*" with CAN bus as a method for communication. In particular, the OBD-II standard specifies the OBD-II connector, incl. a set of five protocols that it can run on. Further, since 2008, CAN bus (ISO 15 765) has been the mandatory protocol for OBD-II in all cars sold in the US, which basically eliminates the other 4 protocols over time. The OBD standards are highlighted in [24].

The OBD provides an interface to access the internal sensor readings and fault codes of the vehicle. The vehicles considered in this study are equipped with sensors which measure Engine Load, RPM, speed, CO2 Emission, Manifold Pressure, Throttle, Acceleration, Coolant Temperature and Fuel Flow. Along with these the device collecting the data adds timestamps, Ambient Air Temperature, GPS Co-ordinates, distance travelled and

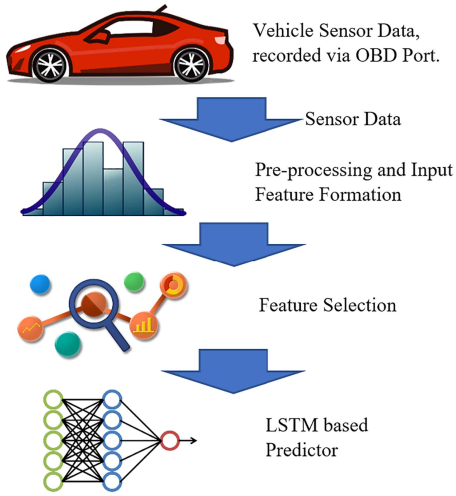

Fig. 4. Architecture of the proposed system for CO2 prediction.

Ambient Pressure. Further virtual features like mileage and fuel consumption are created from distance and fuel flow.

The sensors can have different rates of measurement and values get updated after a fixed interval of time. For this study, the dataset used accesses the OBD once every second.

IV. Proposed Methodology

The proposed model has three stages namely, Feature Sequence Formation, Feature Selection, and LSTM based estimation. Pre-processing and sequencing of data are performed in the Sequence formation stage. Optimal feature subset selection is performed in Feature selection. Finally, LSTM is trained to estimate CO2 emissions using the sequences of data. The details of the stages are briefly discussed. Fig. 4 summarizes the proposed architecture. The proposed model can be deployed on a cloud with the vehicle sending the data at fixed intervals (equal to the window size of the model) and the server using the forward prop of the LSTM to reply with the estimated CO2 emission of the vehicle. The sampling rate of such an architecture will be $1/(Interval\ Length)$.

A. Feature Sequence Formation

1) Pre-Processing: The data was normalized to ensure all features are on the same scale. Further, regex based parser were used to convert string data to float values and strip units and commas. Normalization ensures smooth training and not normalizing might also cause vanishing gradient problem while training out LSTM Network. Min-Max normalization has been used for normalizing the data. The formula for min-max normalization is

$$y_i^j = \frac{x_i^j - \min(x)}{\max(x) - \min(x)} \tag{6}$$

where x_i^j is the j^{th} feature value of the i^{th} input sample and y_i^j is the corresponding normalized value.

2) Sequence Formation: Since vehicle telematics is a time series based data, we cannot directly train a model treating each timestamp as a feature vector. An LSTM model is instead trained on a sequence of such instantaneous features to allow the model to learn the sequential relationships of the data. For obtaining this sequence a moving window of width 64 has been used with an overlap of 50%. The width of the window will affect the feature-length and consequently the model performance. The width of 64 is chosen by experimental analysis. The feature sequences are normalized to avoid the vanishing gradient problem while training the LSTM network. Fig. 5 shows the overlapping sliding window-based sequence generation procedure. Each rectangular box in the figure represents a window. It can be seen that each window captures 64 time steps and has a 50% overlap with the next window. Each time step in our case is 1 s. Hence, each window contains OBD data of 64 seconds length. For Example Window-1 contains data from 0–64 seconds, Window-2 contains data from 32–96 seconds and so on.

TABLE I
Description of Feature Set

Feature	Units	Range	ID
Engine Load	%	0 - 100	1
Speed	km/h	0 - 121	2
Coolant Temperature	$°C$	60 - 100	3
Engine RPM	$Revolutions/min$	0 - 5500	4
Mileage	km/l	0 - 45	5
Fuel Flow	cc/min	0 - 350	6
Throttle	%	0 - 100	7
Acceleration	km/s^2	-25 - 45	8

B. Feature Selection

Features directly affect any estimation and are thus very important. The presence of redundant and irrelevant features negatively affect the performance of the model. To overcome this, Feature selection has been done using co-relation values and Principal Component Analysis (PCA) [25] loading scores.

PCA maps features to a lower-dimensional space by transforming the variables to a new set of variables, that are orthogonal. It attempts to reduce the dimensionality of data, conserving maximum variance. In this study, loading scores of PCA have been used as a ranking method for feature selection.

Fig. 6 shows the biplot of the features with the first two principal components (PC), which have a combined variance score of 84%. PCA generates a loading matrix, each row of which are the weights of the various feature for that component. A biplot shows loading values with a rotated co-ordinate system such that the x-axis is aligned with PC-1 and the y axis is aligned with the PC-2. The length of the vectors of different features on this plot are proportional to the contribution of the feature in the PC. The features are ranked based on their vector-lengths in the biplot. This technique for feature selection has also been used by authors in [26]. Var 1-8 in the plot are feature IDs as given in Table I. It can be seen from the biplot that Speed, Engine RPM, Mileage, Fuel flow, Throttle and Acceleration have the greatest effect on the principal components(PC). Hence, these have been used for estimating CO2 emission in this study.

Feature correlation carries a lot of relational information of features which can be useful in predicting missing values. Fig. 7 shows the feature correlation matrix. It can be seen that the selected features have low co-relation scores, showing that the features add new information to the set.

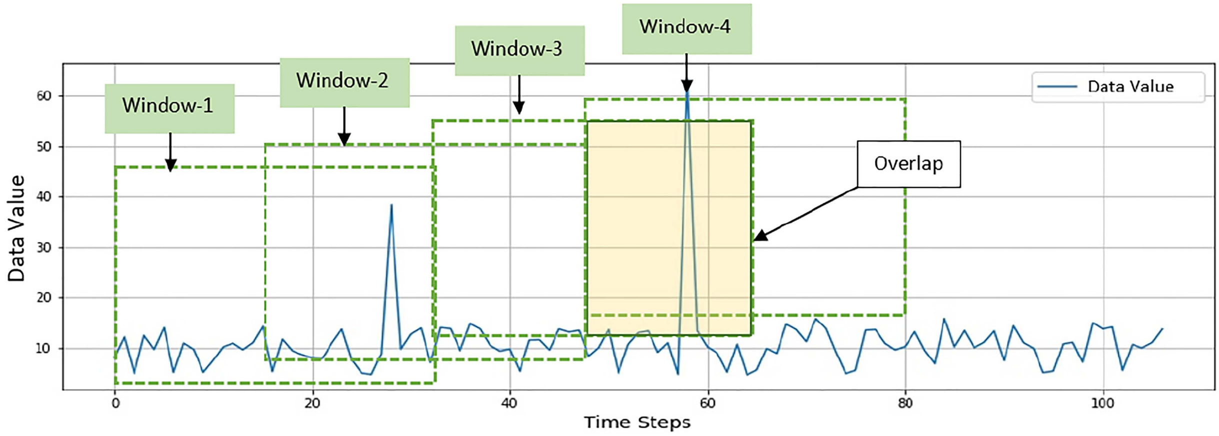

Fig. 5. Illustration of the sliding window feature sequence generator.

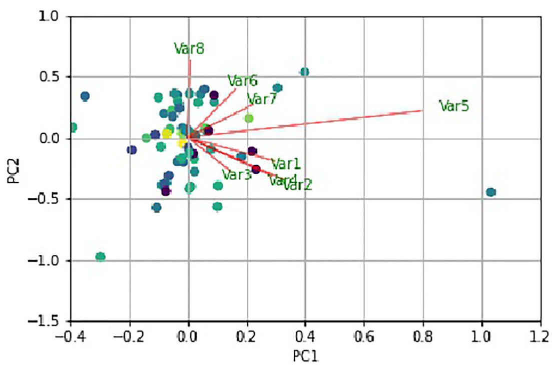

Fig. 6. Biplot of the PCA using the first two components.

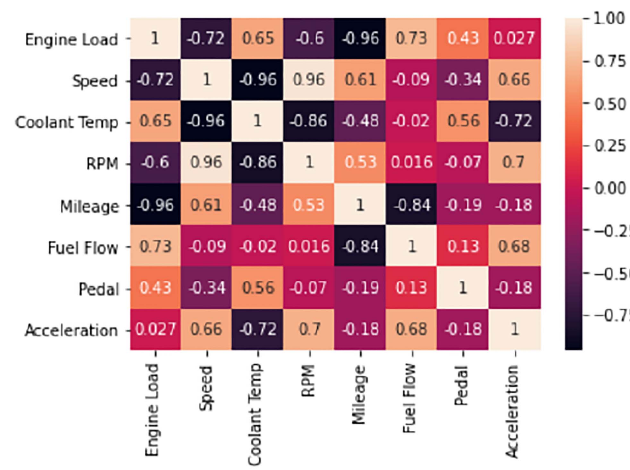

Fig. 7. Correlation matrix of vehicle telematics feature set.

TABLE II
SUMMARY OF DATASET FEATURES

	Vehicle 1	Vehicle 2
Engine	1.0 16v	1.6 16v
Max RPM	7000	7000
Engine	1.0 16v	1.6 16v
Transmission	5	5
Power	76cv	122cv
Manufacturer	Renault	Hyundai
Model	Sandero	HB20%
Engine Type	Petrol	Petrol
Trips	36	8
Trip time	28 hours	3 hours
Nature of Trip	Naturalistic	Controlled
Drivers	10	4
Gender	6M - 4F	2M - 2F
Age	25-61	20-53

C. LSTM Based Predictor

The generated input samples are in the form of sequences and normal machine learning techniques like Support Vector Machine (SVM) and standard feed-forward Neural Networks cannot be used for prediction. Unlike standard neural networks which are feed forward in nature, LSTM has connections for feedback, which allow it to process sequential inputs remarkably well. In this study, a three layer LSTM network has been proposed to predict the CO_2 emission(g/km) of an input.

The hyper-parameters of a model like the number of layers and nodes directly affect its performance and complexity. To find the optimal setting a random search using training and validation data has been performed. A three layer model with 120, 240 and 500 neurons in each layer respectively is found to be the optimal network architecture. Dropout Layers with a dropout rate of 0.2 have been added between the LSTM layers to ensure that all the nodes are trained equally. The model is trained using an RMS Prop Optimizer [27] with Mean Squared Error (MSE) Loss for 50 epochs.

LSTM networks are well suited for time series classification since they are insensitive to lags and gaps of unknown duration in the data. LSTM networks are generally preferred over RNNs because RNN is prone to vanishing gradient problem. The gap insensitivity of an LSTM gives it an advantage over RNNs and Markov based predictors.

V. DATA AND EXPERIMENTATION

A. Dataset

For training and evaluation of the model, an open-source Vehicle Telematics Dataset collected and compiled by Dr. P. Rettore, Computer Science Department, Federal University of

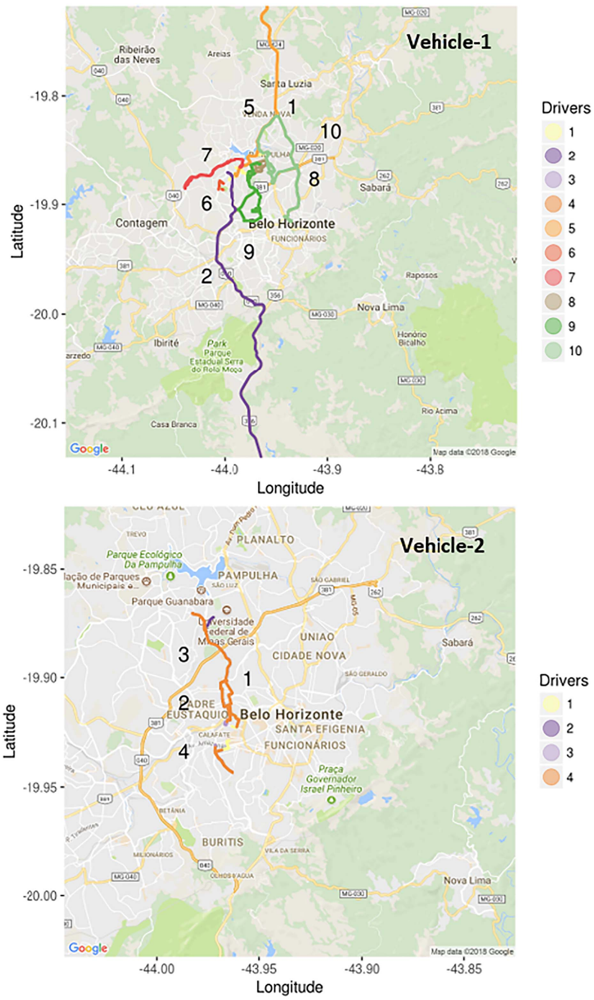

Fig. 8. Routes taken by the different drivers for data collection. Image taken from [11].

Minas Gerais is used [11]. The details of the data collection including sensors and adapters used, vehicles considered and route taken by drivers is outlined in [12], [28]. The composition of the dataset has been briefly explained below. The dataset is prepared using data collected from 14 drivers and 2 vehicles. All four drivers sharing vehicle 2 were asked to drive through 2 different routes while the 10 drivers of vehicle 1 followed their daily commuting routes. For driver's privacy the start and end of the trips were deleted. Fig. 8 shows the route taken by the drivers for both the vehicles. The images have been taken from [11].

The authors used the OBD-II interface as the means of accessing the vehicle's data, transferring them via the Bluetooth connection to a smartphone with the Android operating system, where the data is processed and registered through an app. The sensors are observed once every second. The sensors monitored by the authors also includes an instantaneous CO2 emission measurement sensor. The readings from the CO2 sensor are used as true values for evaluation. The CO2 sensor's accuracy and validation is summarized in [28], where authors propose a low emission driving strategy based on the reduction in CO2 emission observed. Table I summarizes the dataset description.

TABLE III
SUMMARY OF DATASET SPLIT

	Vehicle	Driver
Training Set	1	1 to 8
Validation Set	1	9 to 10
Testing Set	2	1 to 4

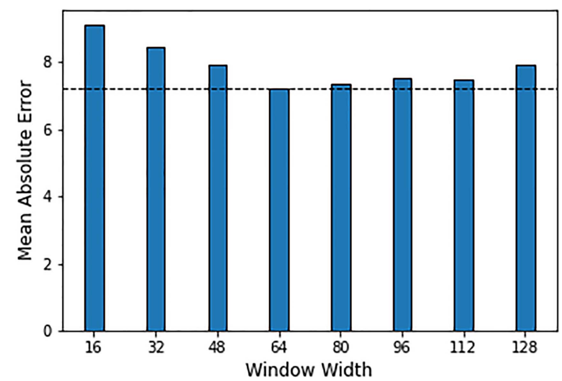

Fig. 9. Comparison of various window widths on the basis of mean square error on the validation set.

Data from Vehicle-2 was used exclusively for testing while the data from drivers 9 and 10 of Vehicle 1 form the validation set. The Training and Testing split is summarized in Table III. Since the study uses open-source data, it becomes important to address some of its limitations. Firstly, the data contains logs from only 2 vehicles which creates a size limitation which is primarily due to the difficult OBD data collection procedure. Secondly, the data is collected only for normal vehicle behavior and cannot be used easily in diagnosis/health based estimations.

B. Experimentation

Various experimentation have been carried out on the model and a brief description of the results obtained is presented. The results of the experimentation is on the Validation Set samples.

1) Comparison of Window Widths: The proposed model first generates feature sequences from continuous data using the sliding window technique. Window width will determine how much information is being passed to the LSTM model. Different window widths have been compared on the basis of their performance on the validation set. Fig. 9 shows the mean squared error with different window widths calculated on the validation set. It can be seen that 64 window length provides the least error and is hence chosen.

2) Comparison of LSTM Structures: The number of layers affects the complexity of an LSTM network. Increasing layers may adversely affect the performance of the model since the model works optimally only at the ideal depth. Different LSTM architectures have been compared in this study namely Single Layer LSTM, DOuble Layer LSTM, Triple Layer LSTM and 4-Layer LSTM. The comparison is based on the performance of the models on the validation set. All training is done on the test set. Table IV shows the result of the comparison of network architectures. It can be seen that the three-layer architecture works best as found by the random-search.

TABLE IV
SUMMARY OF COMPARISON OF LSTM DEPTHS

LSTM Structure	MAE	RMSE
1-Layer	11.45	13.33
2-Layer	9.88	11.94
3-Layer	7.23	10.26
4-Layer	8.04	10.98

TABLE V
COMPARISON OF RESULTS USING DIFFERENT FEATURE SETS

Feature Subset	MAE	RMSE
4 Features	12.48	15.27
6 Features	7.23	10.26
8 Features	9.67	12.13

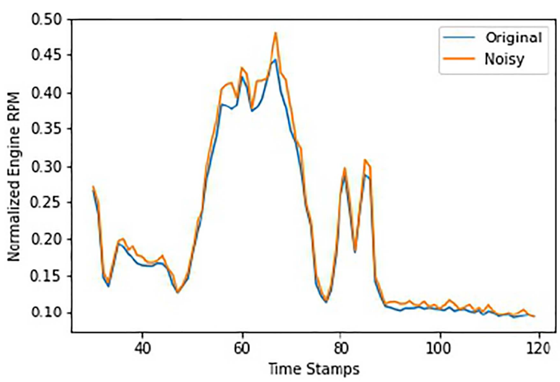

Fig. 10. Normalized Engine RPM actual values with noise added values.

3) Comparison Without Feature Selection: The improvement provided by the selection of features has been highlighted in this section. The various feature sets considered are as follows-

- *4 Features* - Speed, Acceleration, Fuel Flow, Mileage
- *6 Features (Proposed)* - Speed, Engine RPM, Mileage, Fuel Flow, Throttle and Acceleration
- *8 Features* - Load, Speed, Coolant Temperature, Engine RPM, Mileage, Fuel Flow, Throttle and Acceleration

The 4 and 6 feature sets were formed by selecting the best 4 and 6 features respectively using PCA as described in Section IV-B. The results of the comparison have been summarized in Table V.

4) Performance Against Noisy Data: The data used in this study was collected directly from vehicle OBD-ports under specialized conditions, thus, this data is free from noise and other perturbations. In real life, however, when this data will be transmitted using Low-Power Wireless Personal Area Network (LoWPAN) or other low power networks, the data will pick up some noise. To test the model against such noisy data, random noise was added to the data values. For every value, a random positive bias of up to 5% has been added with a 10% probability. Fig. 10 shows a sample of actual Engine RPM data with a noisy version of the data, The result of the analysis with noises has been discussed in the result section.

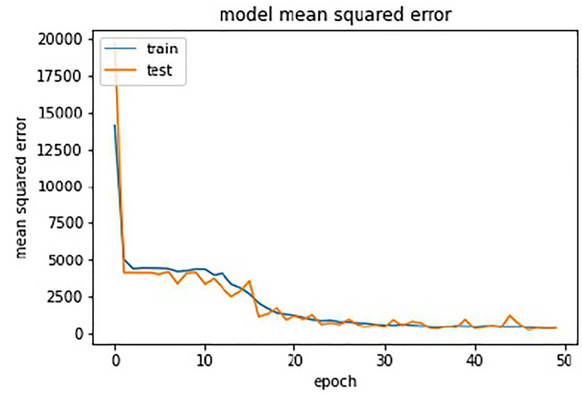

Fig. 11. Mean Squared Error Loss while training the LSTM Network.

VI. RESULTS AND DISCUSSION

The evaluation metrics used are, Mean Squared Error (MSE), Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The formulas are given below

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2 \tag{7}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i) \tag{8}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2} \tag{9}$$

where x_i and y_i are the i^{th} element in x and y.

The proposed methodology is implemented on python software, run on a CPU. The system architecture uses an Intel Core i7 processor with a 4 GB graphic card, 64-bit operating system running at 1.80 GHz, and 16 GB RAM.

As discussed in Section IV-A the proposed model is evaluated on the test set which only contains samples of vehicle-2. The model has not seen any data related to vehicle-2 while training. Thus, using vehicle-2 ensures that the testing is robust and that the model does not overfit. Further, this shows that the model's prediction is independent of the model and make of the vehicle. Fig. 11 shows the training curve of the proposed model with MSE loss.

The model achieves an MAE of 6.16, an MSE of 86.45 and an RMSE of 9.30. The results achieved are very promising considering that the model has been evaluated on a new vehicle which it has never seen. Such a model can be applied to a large number of vehicles irrespective of their make or model. A large scale application of this is thus possible since a single model is capable of predicting vehicle emissions with only general characteristics of the vehicle like engine capacity, size, etc.

For comparing the model against other Emission prediction techniques, Deep Neural Networks and Deep Convolutional Neural Networks have been used because of their popularity. A 3-layer Deep CNN network and a 4-Layer Deep Neural

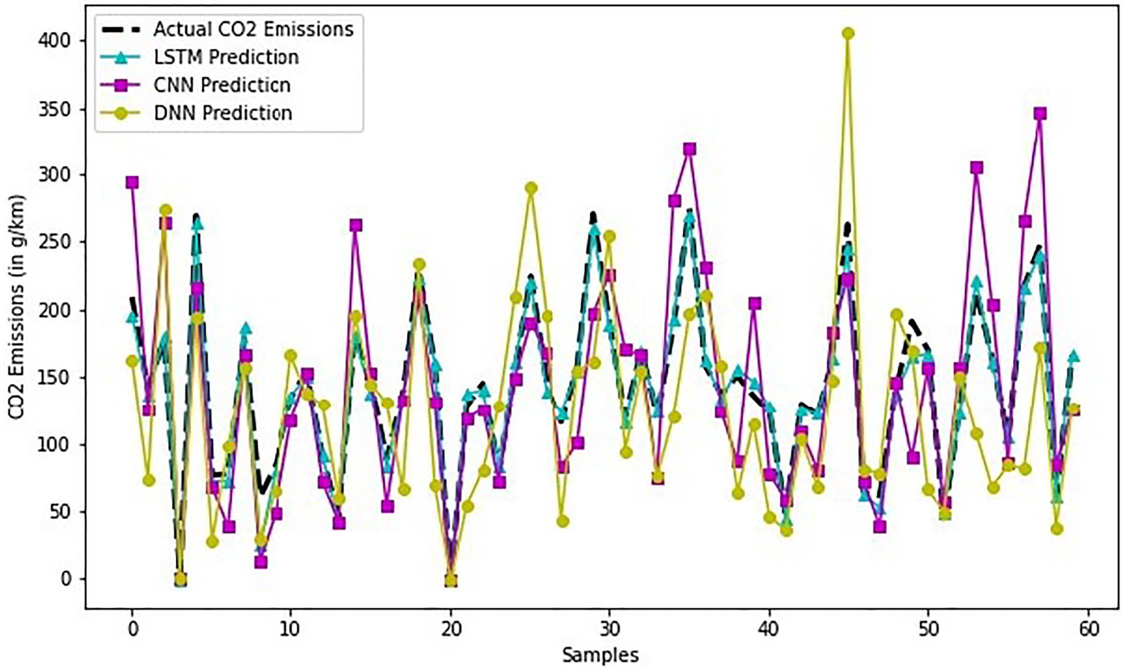

Fig. 12. Comparison of various model's prediction on the test set along with true values.

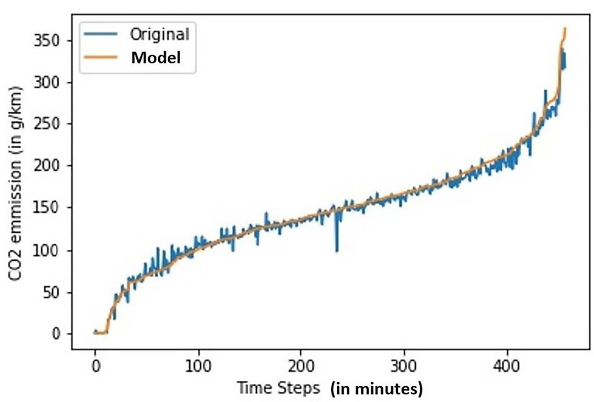

Fig. 13. Comparison of proposed model's prediction on time sequential test set along with true values.

Network have been compared with the proposed method. Fig. 12 compares the performance of various models on the test set. The figure shows the predictions of various models and the true values on the first 60 samples of the test set. Each sample has OBD data of 120 seconds. The samples have been shuffled in the plot. The prediction on un-shuffled test data can be seen in Fig. 13. It can be seen that LSTM performs significantly better than other models.

Further, the performance of the model is evaluated using noisy data to simulate real-life conditions. The summary of the results from noise based test data is summarized in Table VI. The comparison shows that the predictions of CNN and DNN are significantly affected by noise, with errors rising by as much as 20%. LSTM on the other hand seems to be unaffected by noise. Apart from comparison of various deep-learning models, the model has also been compared to standard techniques in the literature, whose analysis is presented in the next section.

A. Comparison to Other CO2 Estimation Techniques

To evaluate the model performance, a comparative analysis has been conducted with respect to other CO2 models available in the literature. The comparison has been done using the RMSE and MAE scores on original as well as noisy test data. The comparisons show that the model developed, outperforms the techniques currently in the literature.

The model has been compared to standard vehicular CO2 estimation techniques. The following models have been considered for the comparative study :-

- *Mileage and Distance-Based Estimation* - This is the most standard CO2 estimation technique used on a large scale to monitor emissions. Simple vehicle mileage and distance product is used to estimate the CO2 emission of the vehicle [9].
- *Speed and Acceleration Based Estimation* - Estimation based on speed and acceleration has been shown to perform well for instantaneous estimation. This has been studied in [8].
- *Proposed Model* - The model described in the study with 6 features for CO2 estimation. Additional features like Fuel Flow, RPM, Mileage and, Throttle are also considered in addition to speed and acceleration.

The LSTM model, defined in Section IV, has been used to estimate CO2 emissions based on the above feature sets. The results of the comparative analysis is summarised in Table VII.

TABLE VI
SUMMARY OF RESULTS OF PROPOSED METHOD COMPARED AGAINST CNN AND DNN ON NOISY AND ORIGINAL TEST SETS

	Deep CNN	Deep NN	LSTM (Proposed)
RMSE (Original Data)	17.82	64.87	9.30
RMSE (Noisy Data)	37.46	124.51	13.15

	Deep CNN	Deep NN	LSTM (Proposed)
MAE (Original Data)	17.98	47.53	6.16
MAE (Noisy Data)	45.48	89.67	11.36

TABLE VII
COMPARATIVE ANALYSIS OF TECHNIQUES

	Mileage and Distance	Speed and Acceleration	Proposed Model
MAE (Original Data)	38.98	10.23	6.16
MAE (Noisy Data)	51.76	15.87	11.36
RMSE (Original Data)	87.45	12.54	9.30
RMSE (Noisy Data)	106.67	21.45	13.15

VII. CONCLUSION

The study describes a new OBD-II data-based technique for vehicular CO2 emission monitoring. The designed system has been tested with noise added data for more practical testing. The high performance given by the model shows that it is highly robust and that OBD-II based systems can be used for estimation of vehicular emissions. The major contributions from this study are briefly outlined below:-

1) A Robust LSTM based deep learning model for continuous CO2 prediction from OBD-II data. The model uses OBD-II sensor data to learn vehicle emission characteristics.
2) An experimental analysis on effects of sensor noises on model performance outlining problems with a sensor data-based model. Such a study will allow future researchers to understand the various issues when working with raw OBD-II data.
3) Comparative study on the effect of varying model structures and window widths. The experimentation ensures the model is optimal.
4) Comparative analysis with other popular deep learning models like Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN).

OBD-II data can give very valuable insights into the vehicle performance, like its health, emission characteristics, etc. Further, in-vehicle sensor data contain knowledge on the vehicle emissions and can be used for accurate CO2 estimation. The outlined method performs better than speed and acceleration based emission estimators in terms of performance and scalability because it takes into account a wider array of sensor data.

The environment of the vehicle including traffic, road-conditions etc. will have an effect on the OBD-II readings and will ultimately also affect the CO2 estimations of the model. This allows the model to be susceptible to changes in the vehicle's environment. The model can be used for vehicular CO2 emission estimation at very large scales. With OBD capturing real-time data, such a system will allow real-time continuous monitoring of emissions.

The study uses an open-source dataset for estimating CO2 from vehicle OBD-II data. One limitation of the study that needs to be addressed is the size of the dataset. A thorough analysis of the proposed methodology on different types of vehicles and different conditions is needed to better understand the outcomes of predictive techniques on vehicular data. The designed system learns the feature patterns to estimate emissions and thus in theory generalizes well to different types and classes of vehicles. This however has not been verified and the future works should focus on validating models on vehicles of different sizes and types.

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